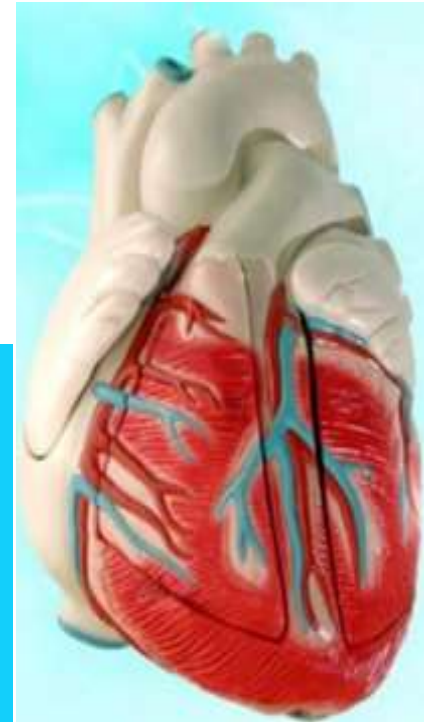




BIOCHEMISTRY OF CVS MODULE

LIPOPROTEINS METABOLISM AND HYPERLIPIDAEMIA

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DEC.,2023
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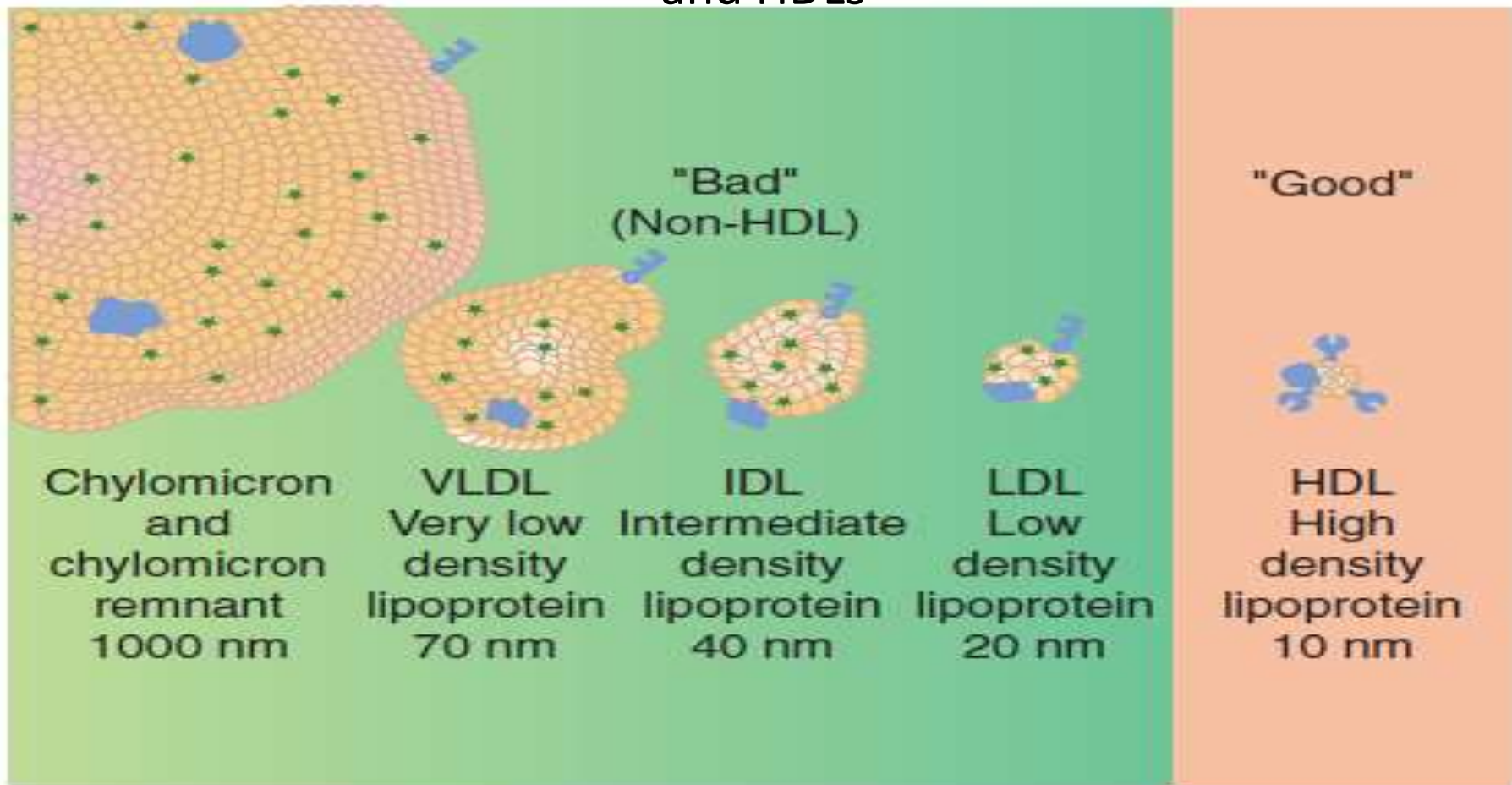
Objective

- Definition of lipoprotein
- Source of lipoproteins
- Classification of lipoproteins based on their size, lipid composition and the type of protein
- Functions of apolipoproteins or apoproteins
- Lipoproteins metabolism and hyperlipidaemia
- Metabolism chylomicrons, VLDL, HDL, LDL, and HDL
- Determination of lipid profile: TAG, VLDL, LDL, HDL, TC

Lipoproteins

- Since lipids are insoluble in water, they are transported in the plasma as protein–lipid complexes (lipoproteins)
 - Or water insoluble lipid to become soluble in plasma
- They are differ based on their
 - size
 - lipid composition and
 - the type of protein they contain
- ❖ The proteins embedded in the lipoproteins have role of transporting lipids through the bloodstream to tissues
 - ✓ where they are Used for energy, stored or to synthesis hormones

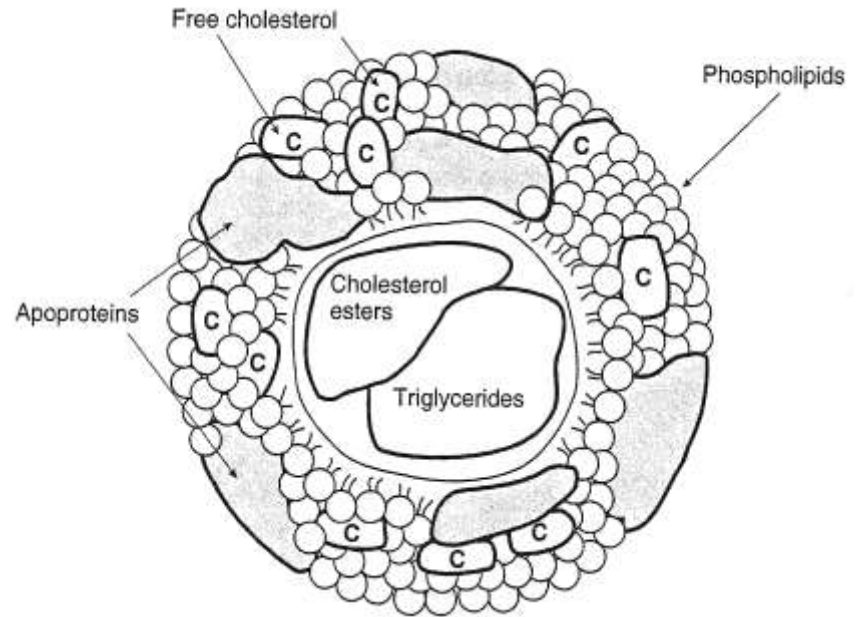
Classification of lipoproteins are chylomicrons, VLDLs, IDLs, LDLs, and HDLs



Fat has a density of about 0.88 g/ml, whereas protein has a density of about 1.0 g/ml as consequence, the higher the fat content, the lower the density of the particle

Plasma Lipoproteins Structure

- The apolipoprotein B (apoB) is essential for initiating the assembly of triglyceride rich lipoproteins in enterocytes and hepatocyte
- LP core
 - Triglycerides
 - Cholesterol esters
- LP surface
 - Phospholipids
 - Proteins
 - Cholesterol



Plasma Lipoproteins Structure...

- Enterocytes edit the mRNA for apo-B at 48% of the length
 - to produce apo B48
 - w/c assembly of the chylomicron (CM)
- Both the CM and VLDL transport **triglyceride** and smaller amounts of cholesterol
- Each chylomicron contains several different apoproteins besides apoB-48,
 - eg apoA-I, apoA-II and acquires apo C-II and apoE

Functions of apolipoproteins

The apoproteins

- a. it combine and transport lipids, to add hydrophilicity and structural stability of the lipoprotein particle
- b . They activate certain enzymes required for normal lipoprotein metabolism
 - ✓ **Apo CII** activates lipoprotein lipase LPL and apo C-III inhibits it.
 - ✓ **Apo AI**: structural component of HDL, w/c activates **LCAT**
- c. To recognize the lipoprotein receptors in the liver and peripheral tissues

Apolipoproteins...

- ApoE is an important apoprotein for the clearance of remnants of the triglyceride-rich lipoproteins
- Apo-100 is an important apoprotein for the clearance of remnants of the cholesterol rich lipoproteins
- Cholesterol ester transfer protein (CETP) exchanges neutral lipids between lipoproteins,
 - causing a net movement of cholesterol ester from
 - HDL to CM and VLDL and
 - a net movement of triglyceride in the reverse direction.

lipoprotein

Lipoprotein	Made in:	Contains mostly:	Function
Chylomicron	Enterocytes (SI)	TG	Deliver TG from SI to cells
Very low density lipoprotein (VLDL)	Liver	TG	Deliver TG from liver to cells
Low density lipoprotein (LDL)	From VLDL*	Cholesterol	Deliver cholesterol to cells
High density lipoprotein (HDL)	SI & liver	Protein	Pick up free cholesterol from cells to take back to liver

Metabolism Chylomicrons

Chylomicrons

- The largest of the lipoproteins and the least dense
- Transports exogenous (dietary) triglycerides to cells
- 90 % by weight is triglycerides
- Absent from fasting plasma, it is removed from the plasma within 6 hours by the liver
- package TAGs together with proteins & phospholipids in chylomicrons (CM)
 - w/c are lipoprotein particles that do not
 - coalesce in aqueous solutions
 - Chylomicrons also contain cholesterol & fat-soluble vitamins
 - The major apoprotein associated with CMs apoB-48

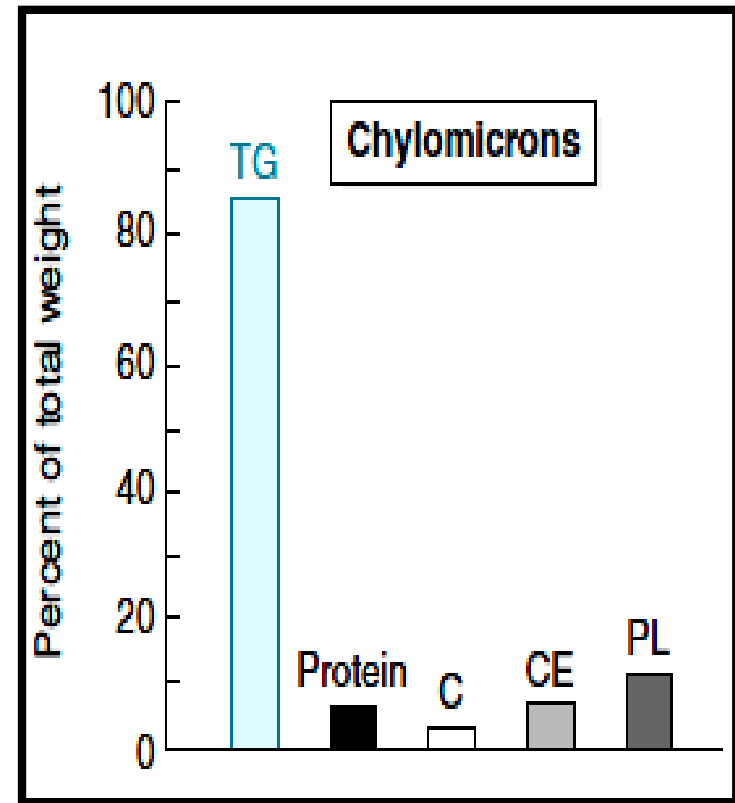
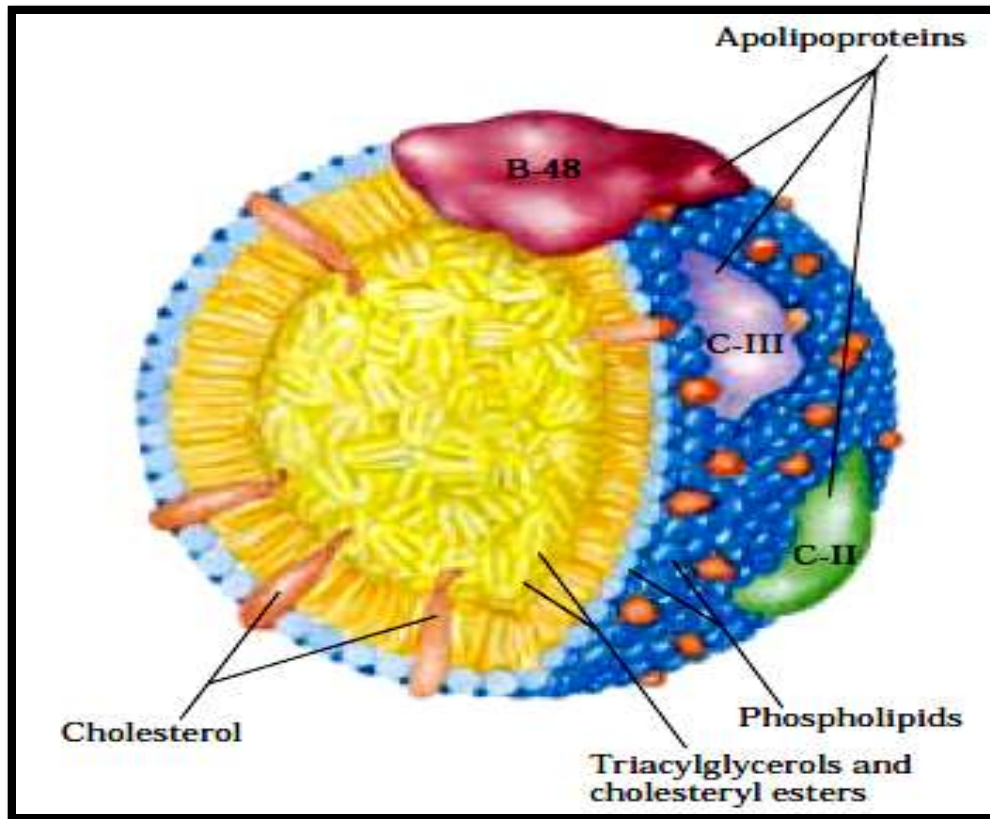


Fig : Molecular Structure & Composition of a Typical Chylomicrons

Metabolism Chylomicrons con't...

- By the process of exocytosis, CMs are:
 - » Secreted by the intestinal epithelial cells
 - » Into the lymphatic system & enter the blood circulation
- Initially, the particles are called nascent CMs
 - As they accept proteins from HDL within the lymph & the blood;
 - » they become “**mature**” CMs
- Major apoproteins of chylomicrons are apoB-48, apoCII, and apoE

Metabolism Chylomicrons con't...

► The TAGs of the CMs are digested by

□ **LPL:**

– Attached to the membranes of endothelial cells that line the capillary walls of

- ✓ adipose cells,
- ✓ muscle cells (particularly cardiac muscle), &
- ✓ cells of the lactating mammary gland

□ The **isozyme** synthesized in adipose cells has

- a higher K_m for CMs than the isozyme synthesized in muscle cells

❖ **Adipose LPL** is more active after a meal

- when CMs levels are elevated in the blood

Metabolism Chylomicrons con't...

- ▶ Insulin stimulates: the synthesis & secretion of adipose LPL after a meal, when TAG levels increase in blood
- The major fate of the FAs is
 - storage as TAG in adipose tissue or may be oxidized for energy in muscle & other tissues
 - The FAs released from TAGs by LPL are not very soluble in water & soluble in blood by forming complexes with the protein albumin
- The glycerol released from CM TAGs by LPL
 - used for TAG synthesis in the liver in the fed state

Metabolism Chylomicrons con't...

- Some portion of a CM remains in the blood
 - after LPL action , known as a chylomicron remnant
- This remnant binds on hepatocytes receptors
 - recognize apoprotein E ,taken up by the process of endocytosis
- liver lysosomes fuse with the endocytic vesicles,
 - & the CM remnants are degraded by lysosomal enzymes
- The products of lysosomal digestion:
 - e.g., FAs, amino acids, glycerol, CL, phosphate
 - can be reused by the cell or may enter the endogenous pathway

Type I hyperlipidemia (familial hyperchylomicronemia)

Patients with deficiency of lipoprotein lipase or apo C-II show

- dramatic accumulation of TAG- rich lipoproteins in the plasma

Very-Low-Density Lipoproteins (VLDL)

- In the endogenous pathway, the liver produces VLDL particles with apo B-100
 - If dietary intake exceeds the immediate fuel requirements of the liver, the FA is synthesized by the liver
 - newly synthesized triglyceride and
 - a small amount of free or esterified cholesterol, in association with phospholipids, and apo-proteins, form VLDL
 - it transports endogenous triglycerides from liver to peripheral tissues
 - 55 - 65 % by weight is triglycerides

Very-Low-Density Lipoproteins (VLDL)

- TAG of VLDL is degraded by lipoprotein lipase (as chylomicrons) as it pass through the circulation
- Its structure is altered as TAG is removed by LPL
- These particles deliver fatty acids to peripheral tissues causing the VLDL
 - to decrease in size and become more dense
- These fatty acids removed are oxidized as fuel by muscle cells
- in fat cells the FAs is used in the resynthesize of triacylglycerols or milk production in lactating breast

Metabolism of very low density lipoproteins(VLDLs)

- Removal of the triglyceride fraction from the particles, while retaining the cholesterol component, results in
 - ✓ their conversion into intermediate density particles and ultimately LDL particles, laden with cholesterol ester

LIPOPROTEIN RECEPTORS

RECEPTORS

A. The LDL Receptor

B. LDL Receptor-Related Protein (LRP)

C. Macrophage Scavenger Receptor

A. The LDL Receptor

- **It mainly supply cholesterol to the extrahepatic tissues.**
 - **LDL receptor, specifically recognizes apo B-100 and apo E**

Therefore, this receptor binds IDL, and chylomicron remnants in addition to LDL.

Metabolism of Low Density Lipoprotein

LDL particles

- retain **apoB –100** , but loose their other apolipoproteins to HDL
- contain much less TAG than their **VLDL**
- have a high **concentration of cholesterol** and **cholesteryl esters**
 - **Why?**
- Synthesized in the liver
- Most atherogenic lipoprotein ... “ Bad Cholesterol
- These LDL particles are the main carrier of cholesterol to p. cells for incorporation into membranes and steroid synthesis

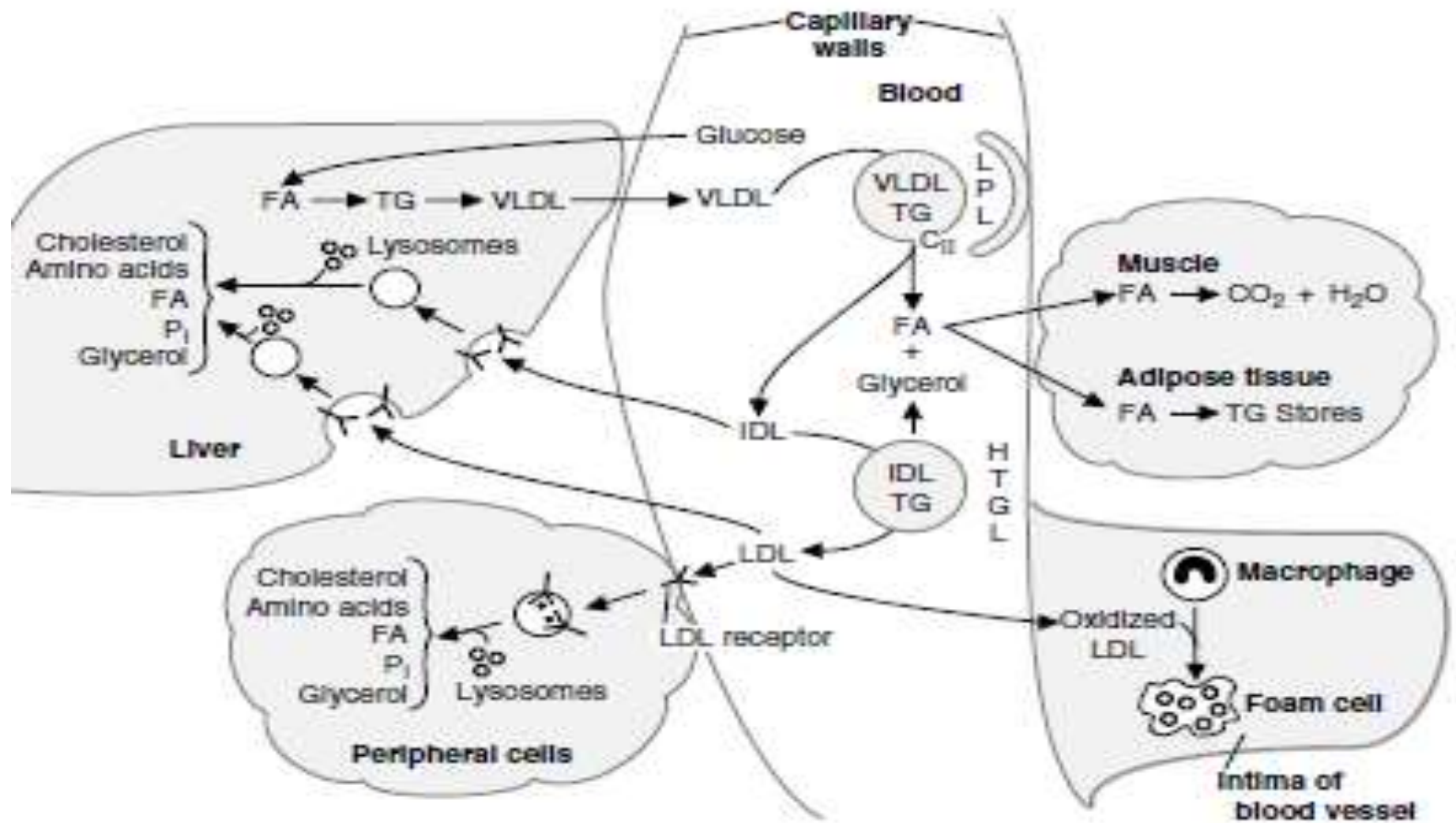


Fig.. Fate of VLDL. VLDL triacylglycerol (TG) is degraded by LPL, forming IDL. IDL can either be endocytosed by the liver through a receptor-mediated process or further digested, mainly by hepatic triacylglycerol lipase (HTGL), to form LDL.

Metabolism of Low Density Lipoprotein

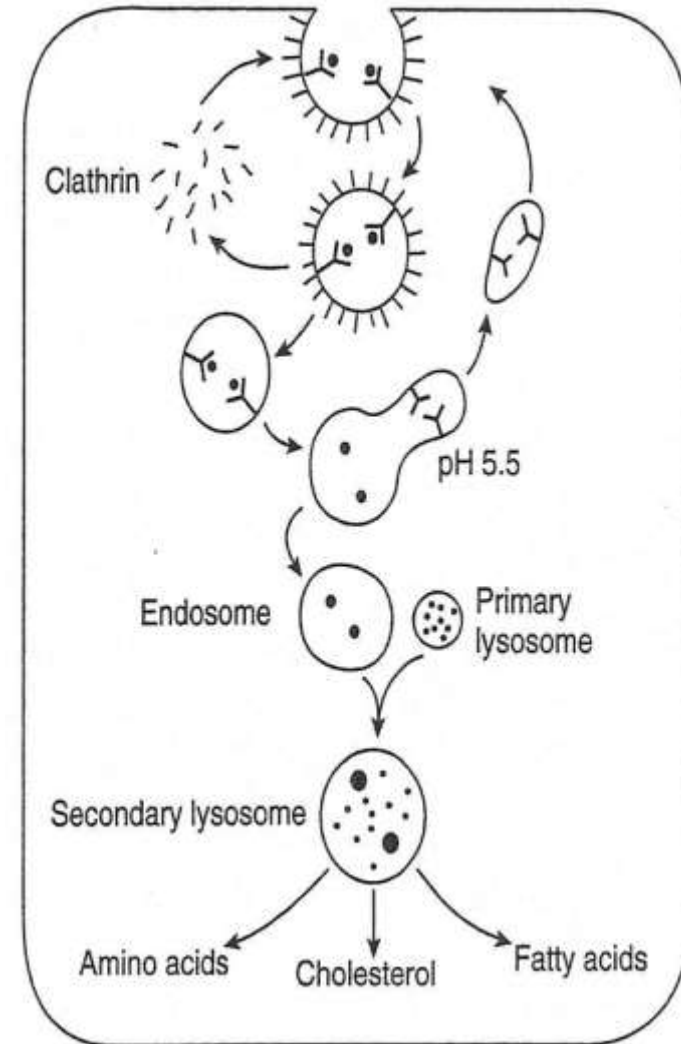
- The primary function of LDL particles to provide free cholesterol to peripheral tissues on the membranes ,
 - by binding to LDL receptors that recognize apolipoprotein B-100
- play a key role in development of atherosclerosis by depositing lipid in the wall of blood vessels

Metabolism of Low Density Lipoprotein

A. A deficiency of **functional LDL receptors** causes a significant elevation in plasma LDL cholesterol, implies plasma cholesterol

✓ w/c lead to coronary heart ds but plasma TAG levels remain normal

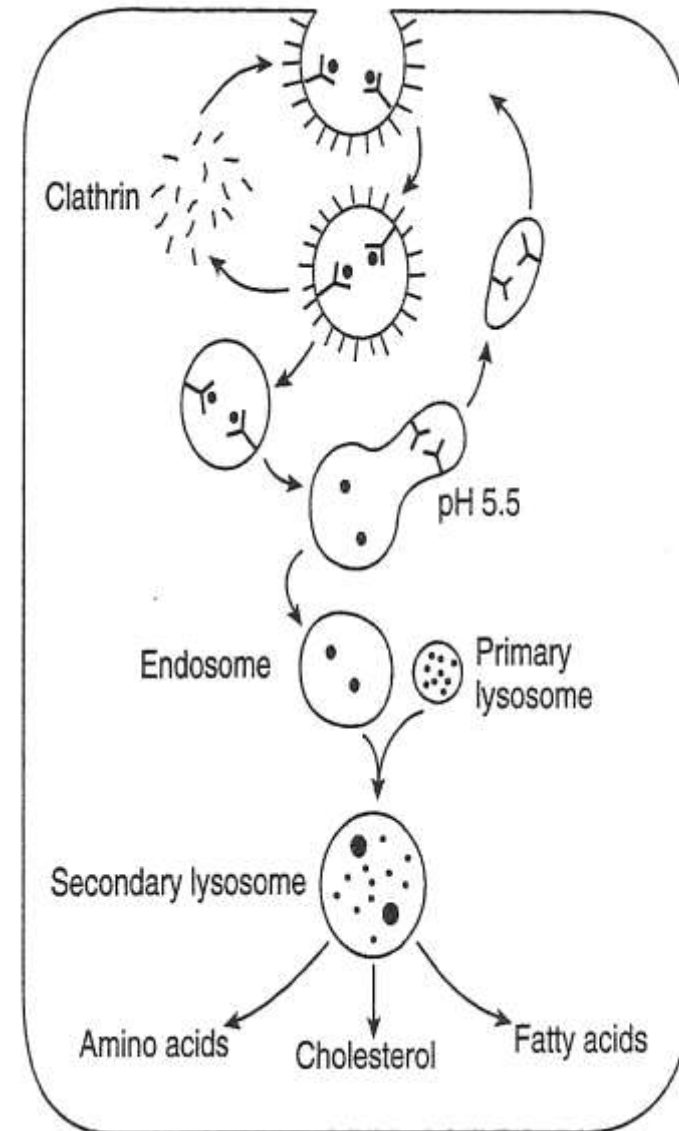
- LDL receptor-mediated endocytosis
- LDL receptors 'Clathrin coated pits'
- Clathrin: a protein polymer that stabilizes shape of pit after endocytosis



Metabolism of Low Density con't...

Loss of clathrin coating is followed by return of receptor to surface
whereas,

- the lipoprotein **remnants** in the vesicle are degraded by **lysosomal enzymes**
- Fusing of endosome with lysosome
 - Free cholesterol, Fats & amino acids



Metabolism of Low Density con't...

- The number of LDL –receptors varies according to the availability of these lipoprotein particles and needs of the cells

For example

- if there is a large amount of circulating plasma LDL
 - ✓ the number of cell- surface receptors for it decreases frequently termed “down –regulation”
- Conversely, if cells are starved for cholesterol they increase the number of cell- surface receptors “up- regulation”
- The higher the LDL level in your blood, the greater chance you have of getting heart disease

Metabolism of Low Density con't...

- Apo-B-100 binding to the LDL receptor results
 - **in internalization of the particle and its removal from plasma.**
- The cholesterol content of the liver cells in turn
 - **regulates the levels of LDL receptors and**
- in order to regulate cholesterol and fatty acid metabolism there are key genes involved
- The genes that are regulated include:
 - HMG CoA reductase which is the rate-limiting enzyme in cholesterol bio-synthesis pathway

Metabolism of Low Density con't...

The regulation of the activity of HMG-CoA reductase is controlled in multiple ways

1.transcriptional control

2.allosteric control of hmg-coa reductase

3.regulation by covalent modification

4. drugs

✓ Statins have their major impact in the liver by lowering cholesterol biosynthesis

➤ **It increased removal of LDL from bloodstream &**

— **by up regulation of LDL-R expression result in the binding apo B-containing lipoproteins**

Metabolism of Low Density ...

- Mechanism for this feedback regulation of cholesterol biosynthesis involves one or more of the SRE-BP
- These proteins , are required for the **full expression** or transcription of the LDL receptor genes, &
 - These Proteins are also capable of sensing the concentration of sterols within the cell
- SREBP1- a specifically enhances transcription of genes required for
 - HMG-CoA reductase expression
- These Proteins or TFs proteins by binding to the sterol regulatory element (SRE)
- upstream of the start of HMG reductase gene, when it bound, the rate of transcription is increased

Metabolism of Low Density ...

- SREBPs, are integral ER proteins ,it is inactive initially
- and the active component of the protein is released by **two proteases ;**
 - SCAP (SREBP cleavage-activating protein) and
 - S2P (site 2 protease)
- Once activated protein is released of ER, the active amino terminal component travels to
 - the nucleus to bind to SREs.
- The soluble SREBPs /TFs are effectively stimulate HMG - reductase mRNA transcription

Metabolism of Low Density ...

- When cytoplasmic sterol levels rise, the sterols bind to SCAP and inactivate it, thereby
- leading to a decrease in transcription of the reductase gene, and
 - less reductase protein being produced

Metabolism of Low Density ...

- When sterol levels are high in a cell, the binding of the **SREBP** to the SRE of these genes is suppressed
- Subsequently synthesis the LDL receptor from **mRNA** is diminished under these circumstances
- This, in turn, appropriately reduces the amount of cholesterol that
 - can enter these cholesterol-rich cells by
 - ✓ receptor-mediated endocytosis or downregulation of LDL receptor synthesis

Metabolism of Low Density ...

- The levels of *LDL receptor* are also regulated by;
 - the secreted proprotein convertase subtilisin /kexin type 9 (PCSK9) w/c synthesized by liver,
- It binds to the LDL receptor and enhances its degradation in lysosomes so
 - Cholesterol level in plasma increase
- PCSK-9 inhibitor block this activities & enhance LDL receptor to internalize LDL-C for utilization or degradation by the liver
- A similar up regulation of LDLRs is noted in the liver
 - **when more cholesterol is required to bile acid synthesis**
 - **When cell starved**

LIPOPROTEIN RECEPTORS con't...

B. LDL Receptor-Related Protein (LRP)

- ✓ LRP is structurally related to the LDL receptor but recognizes a broader spectrum of ligands

C . Macrophage Scavenger Receptor

- This Receptor scavenge
 - cellular debris and oxidized lipoproteins through a range of receptors
- There are a number of different types of scavenger receptors
- SR-B1 is used primarily for HDL binding, whereas
 - **scavenger receptors expressed on macrophages are SR-A1 and SR-A2**

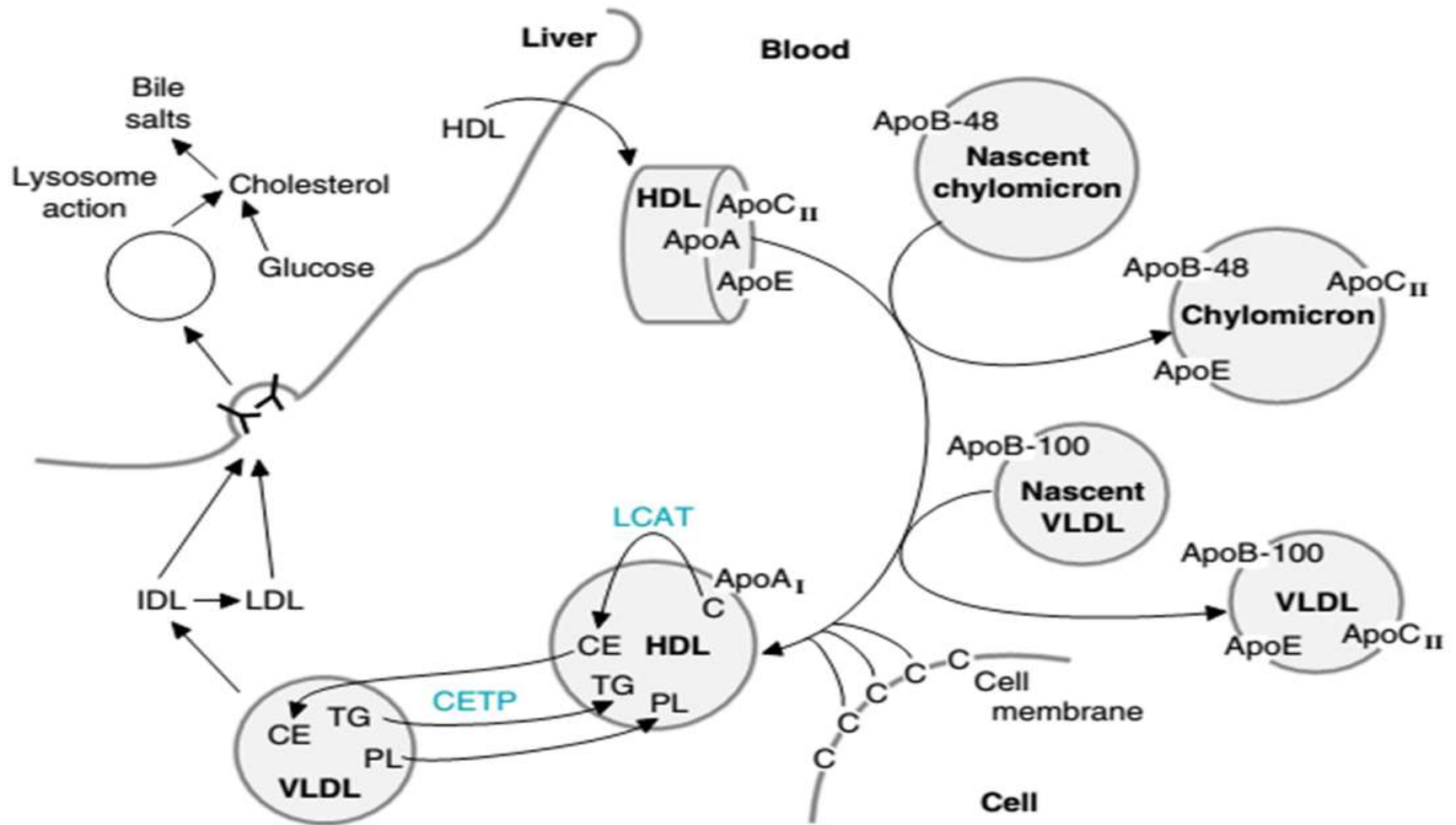


Fig. Nascent HDL is synthesized in liver and intestinal cells. It exchanges proteins with chylomicrons and VLDL. HDL picks up cholesterol (C) from cell membranes

- Some cells, particularly the phagocytic macrophages, have nonspecific receptors known as “scavenger” receptors that
 - bind various types of molecules, including oxidative modified LDL particles

FATE OF HDL CHOLESTEROL

- Mature HDL particles can bind to specific receptors on hepatocytes (apoE receptor), but
- the **primary means** of clearance of HDL from the blood is through its uptake by the scavenger receptor SR-B1
- This receptor is present on many cell types
- SR-B1 receptors can be up regulated in certain cell types, that require **cholesterol for biosynthetic purposes**,
 - such as the cells that produce the steroid hormones
- The SR-B1 receptors are not down regulated when cholesterol levels are high

Metabolism of high-density lipoproteins (HDLs)

■ HDL particles

- The first is synthesis of nascent HDL by the liver and intestine

- Released into the blood stream by exocytosis

They perform a number of important functions:

1. Serve as a circulating **reservoir of apo C-II and apo –E**

2. Removes free cholesterol from extrahepatic tissues and **esterifying** it

- using the plasma enzyme phosphatidylcholine : cholesterol acyltransferase(PCAT) also known as LCAT, where” L “stands for lecithin).

Metabolism of high-density con't...

HDL particles

3. Transfers cholesterol to VLDL and IDL in exchange for TAG by CETP

4. Carries CE to the liver, where the HDL is degraded and cholesterol released

Composition

- 30% phospholipids
- 20% free cholesterol
- 50% apoprotein
 - Apoproteins, predominantly
 - apoA1, apo A-II, apo C- I, apo E and apoC-II

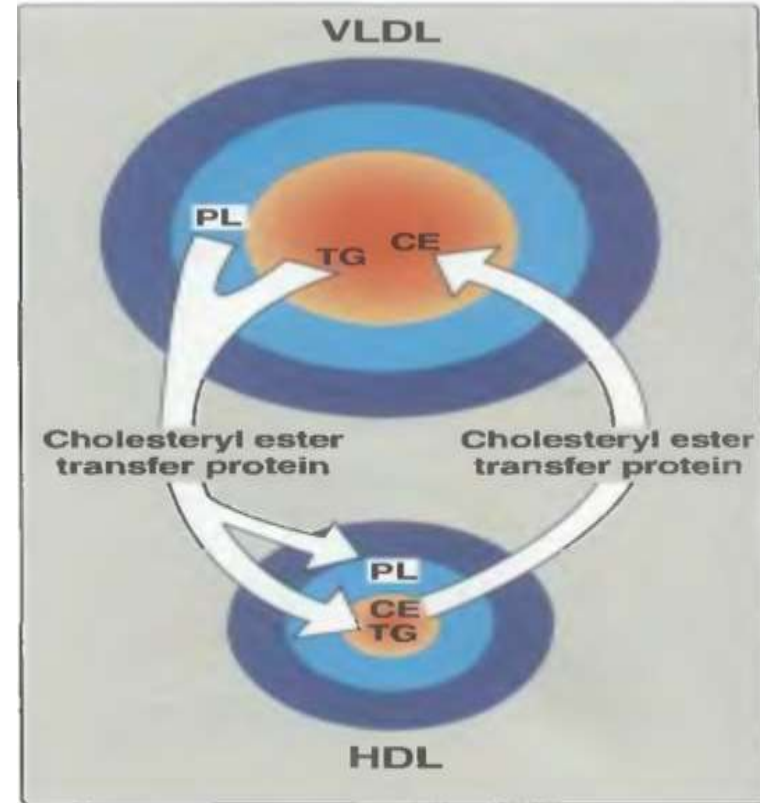


Figure 18.18

Transfer of cholesteryl esters (CE) from HDL to VLDL in exchange for triacylglycerol (TG) or phospholipids (PL).

HDL uptake of free cholesterol

HDL particles are

- An excellent acceptors of unesterified cholesterol from
 - the surface of cell membranes and from other circulating lipoproteins
- Free cholesterol is pick up or exportation from peripheral cell membranes through
 - the adenosine-binding cassette transporter A1 (ABCA1) to HDL particle &
 - esterified by LCAT/PCAT a plasma enzyme synthesized by the liver
- It is activated by apoA-1 of the HDL

Esterification of free cholesterol

- Apoproteins AI - V are found predominantly on HDL w/c serve as structural roles
- The fatty acid from carbon 2 of phosphatidylcholine transferred directly to cholesterol on the C3 to form CE & leaving lysophosphatidylcholine by PCAT
- It is essential for esterification and by doing protection against cholesterol toxicity

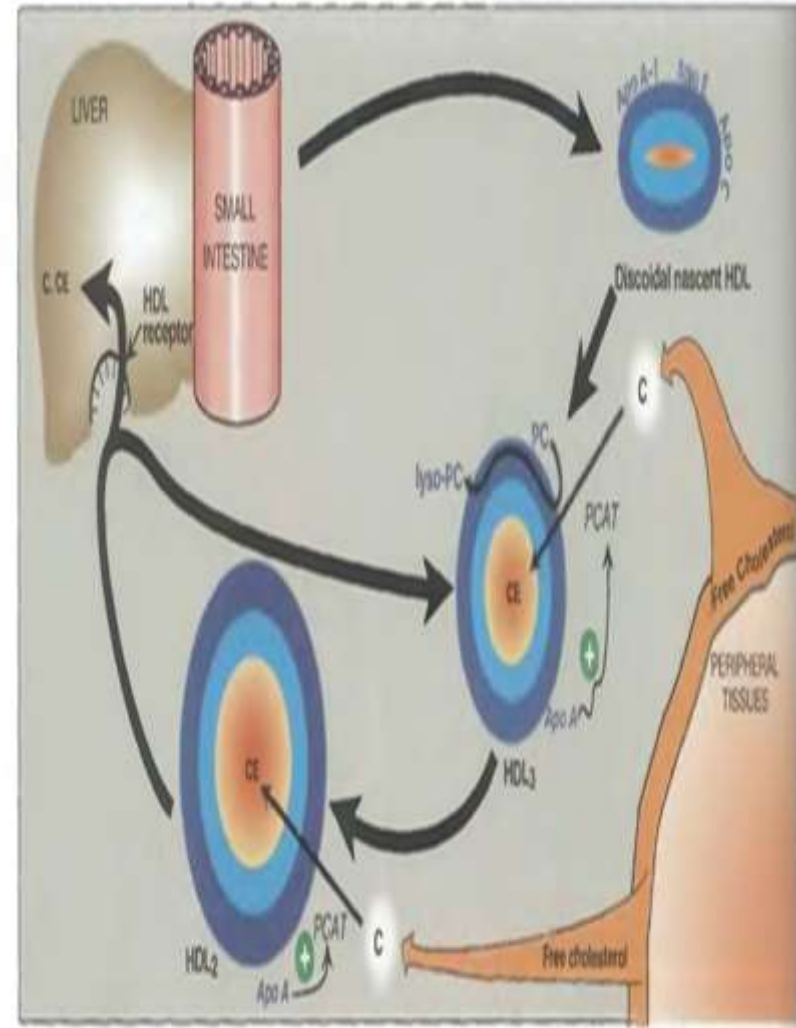


Figure 18.23

Metabolism of HDL. PC = phosphatidylcholine; lyso-PC = lysophosphatidylcholine. PCAT = Phosphatidylcholine cholesterol transferase.

Esterification of free cholesterol

- ❖ HDL particles are taken up by
 - the liver by receptor-mediated endocytosis (apoE receptor), and cholesteryl esters are degraded.
 - **The cholesterol** released can be
 - ✓ either repackaged in lipoproteins, converted into bile acids or secreted into the bile for removal from the body or
 - ✓ to steroido-genic cells for hormone synthesis

➤ . key component of
cholesterol homeostasis

Hyperlipidemia

- Abnormally elevated level of lipid in blood, such as cholesterol and triglycerides
- these lipids can adhere to the walls of the arteries and
 - **restrict blood flow &**
 - **creates significant risk of heart attack and stroke**

There are 3 major lipids in our blood

- Cholesterol, Triglycerides & Phospholipids
 - Cholesterol which is necessary for the synthesis of bile acid
 - Triglycerides : which provides energy to the cell
 - Phospholipids :which is the major component of cell membrane

Hyperlipidemia...

ATHEROSCLEROSIS

- Atherosclerosis is a complex disease characterized by thickening or hardening of arteries due to the accumulation of lipids :particularly free cholesterol, and esterified , collagen, fibrous tissue,
- proteoglycans, calcium deposits etc in the inner arterial wall
- Atherosclerosis is a progressive disorder that narrows and ultimately blocks the arteries

Coronary arteries

- Are the arteries supplying blood to heart if affected leading to
 - **myocardial infarction or heart attacks**

Hyperlipidemia...

Causes of atherosclerosis and CHD :

- The development of atherosclerosis and the risk for the coronary heart disease (CHD) is directly correlated with plasma cholesterol and LDL
- On the other hand, plasma HDL is inversely correlated with CHD

Disorders that may cause atherosclerosis

- These include diabetes mellitus, hyperlipoproteinemias, nephrotic syndrome, hypothyroidism etc.

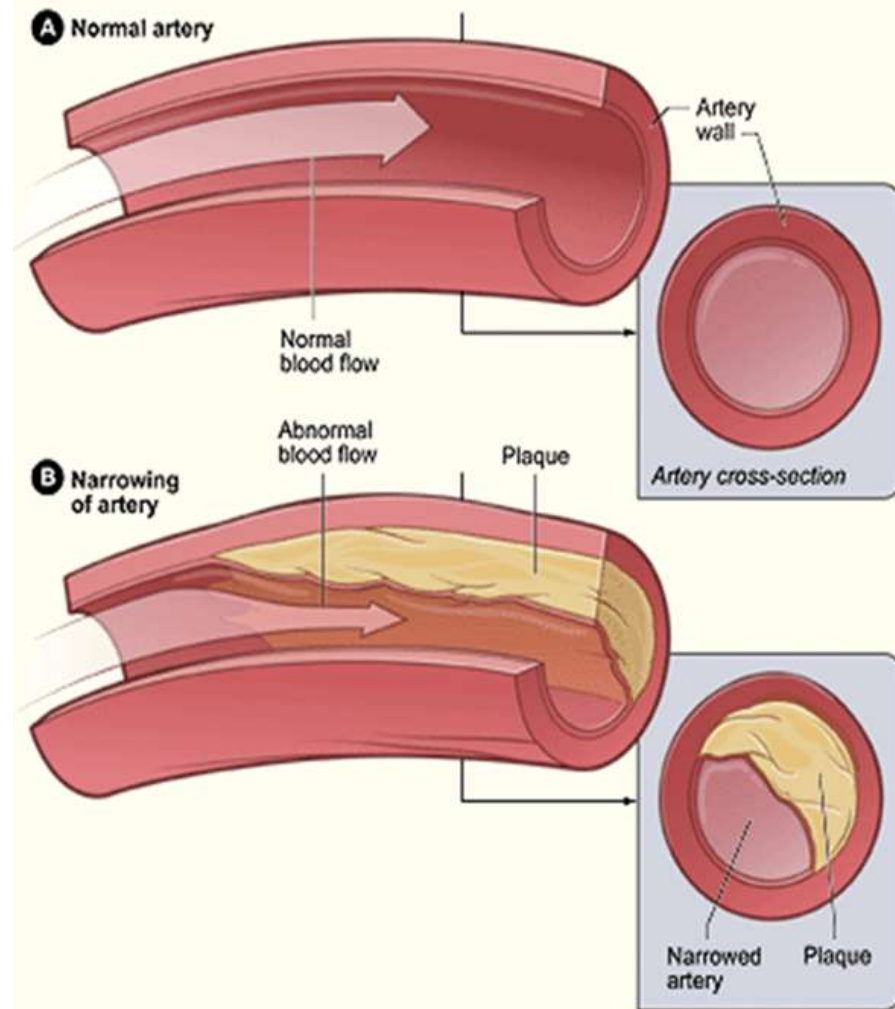


Fig mec.of lipids can adhere to the walls of the arteries and restrict blood flow

Hyperlipidemia...

Relation between HDL and CHD

- The increased levels of plasma HDL (good cholesterol) are correlated with a low incidence of cardiovascular disorders.
- Women have higher HDL and are less prone to heart diseases compared to men
- This is attributed to estrogens in women.
- The following activities increase HDL levels and reduce the risk CHD
 - Strenuous physical exercise & reduction in body weight
 - moderate alcohol intake,
 - consumption of unsaturated fatty acids (vegetable & fish oils)

Lipoprotein-a and CHD

- Lipoprotein a (Lp-a) is almost identical in structure to LDL
- Lp-a contains an additional apoprotein, apo-a.
- Lp-a inhibits fibrinolysis.
- Recent studies have shown that elevation of lipoprotein-a in the plasma (>30 mg/dl) suggests increased risk of CHD.

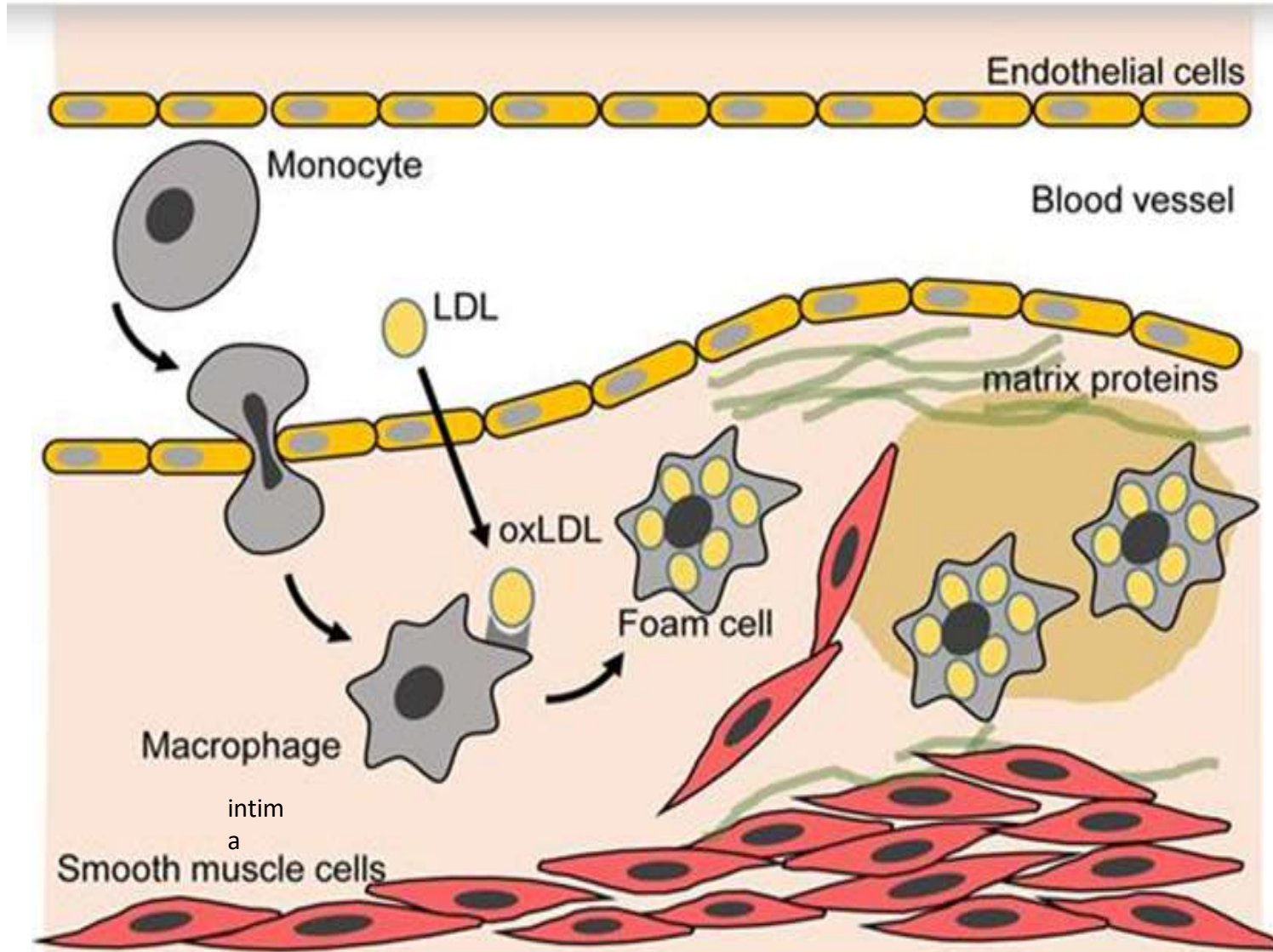
Atherosclerosis process

- Atherosclerosis involves damage to, or dysfunction of, the endothelial cells that form the inner lining of blood vessels
- It resulting in entry of LDL particles into the vessel wall (Fig).
- Lipids and proteins of the LDL particle undergo oxidative modification by reactive oxygen species
 - (e.g. superoxide, O_2^-), oxidized LDL (oxLDL)
- Accumulation of these atherogenic particles (LDL particles) attracts macrophages w/c engulf it ,to form foam cell w/c
 - is give rise to the appearance of 'fatty streaks' within the endothelium

Atherosclerosis...

- this process promote various pro-inflammatory mediators to be produced during this process
- which stimulate smooth muscle cell proliferation, and migration of these cells into the sub endothelial layer
- w/c promoting further the development of atherosclerotic plaques
- The plaque may partially block the lumen of the blood vessel or eventually rupture 'leading to formation of a thrombus
- as blood platelets adhere to the exposed sub endothelial collagen

Atherosclerosis...



Lipoproteins and the process of atherosclerosis

Atherosclerosis...

- Several genetic defects in
 - the LDL receptor and
 - apoprotein genes cause hyperlipidaemia
- Heterozygous familial hypercholesterolaemia is due to mutations in the LDL receptor
- The mutations cause underproduction of the receptor and reduced clearance of the LDL cholesterol from plasma by the liver.
- Mutations in the apoproteins that function as ligands for the LDL receptor (e.g. apo B-100 and apoE) can cause high LDL concentrations and an increased risk of atherosclerosis.

Atherosclerosis...

- LDL-cholesterol can be calculated by the following formula:
- $\text{LDL-cholesterol} = \text{Total cholesterol} - (\text{HDL-cholesterol} + \text{TAG}/5)$
 - Cholesterol is present in plasma mainly in LDL (70%; the “bad cholesterol”) and
 - in smaller amounts in HDL (“the good cholesterol”)

Products Derived from Cholesterol

Cholesterol is not only an essential component of cell membrane, but also a precursor of the following biologically active compounds

1. Vitamin D: Formation of vitamin D₃ occurs by ultraviolet irradiation of the skin.
2. Adrenocortical hormone: The steroid hormones of the adrenal cortex are produced from cholesterol
3. Sex hormones: The male sex hormone, testosterone (19-C), and the female sex hormone, estrogen (18-C) are also derived from cholesterol.
4. Synthesis of bile acids and bile salts

Antioxidants and atherosclerosis

- Antioxidants, in general, decrease the oxidation of LDL.
- There is some evidence, based on the epidemiological studies that taking of antioxidants (vitamins E and C or carotene) reduces the risk of atherosclerosis and CHD.

FATTY LIVER

- The normal concentration of lipid (mostly phospholipid) in liver is around 5%.
- Liver is not a storage organ for fat, unlike adipose tissue. However, in certain conditions, lipids— especially the triacylglycerols
 - accumulate excessively in liver, resulting in fatty liver
- In fatty liver, droplets of triacylglycerols are found in the entire cytoplasm of hepatic cells
- This causes impairment in metabolic functions of liver.

FATTY LIVER....

Fatty liver is associated with fibrotic changes and cirrhosis,
Fatty liver may occur due to two main Causes

1. Increased synthesis of triacylglycerols

2. Impairment in lipoprotein synthesis

1. Increased triacylglycerol synthesis :

- Mobilization of free fatty acids from adipose tissue and their influx into liver is much higher than their utilization
- This leads to the overproduction of triacylglycerols and their accumulation in liver

FATTY LIVER....

- Diabetes mellitus, alcoholism, starvation and high fat diet are associated with increased mobilization of fatty acids that often cause fatty liver.
- Alcohol also inhibits fatty acid oxidation and, thus, promotes fat synthesis and its deposition

FATTY LIVER....

2. Impaired synthesis of lipoproteins :

- The synthesis of very low density lipoproteins (VLDL) actively takes place in liver.
- VLDL formation requires phospholipids and apoprotein B.
- Fatty liver caused by impaired lipoprotein synthesis may be due to :
 - defect in phospholipid synthesis;
 - block in apoprotein formation;
 - failure in the formation/secretion of lipoprotein.

FATTY LIVER....

- LPL or apo CII deficiency will result in extreme disruption of CM clearance, w/c lead diminished VLDL production because of
 - reduced hepatic triglyceride delivery from the gut, and
 - **increased clearance of LDL by up regulation of LDLRs**

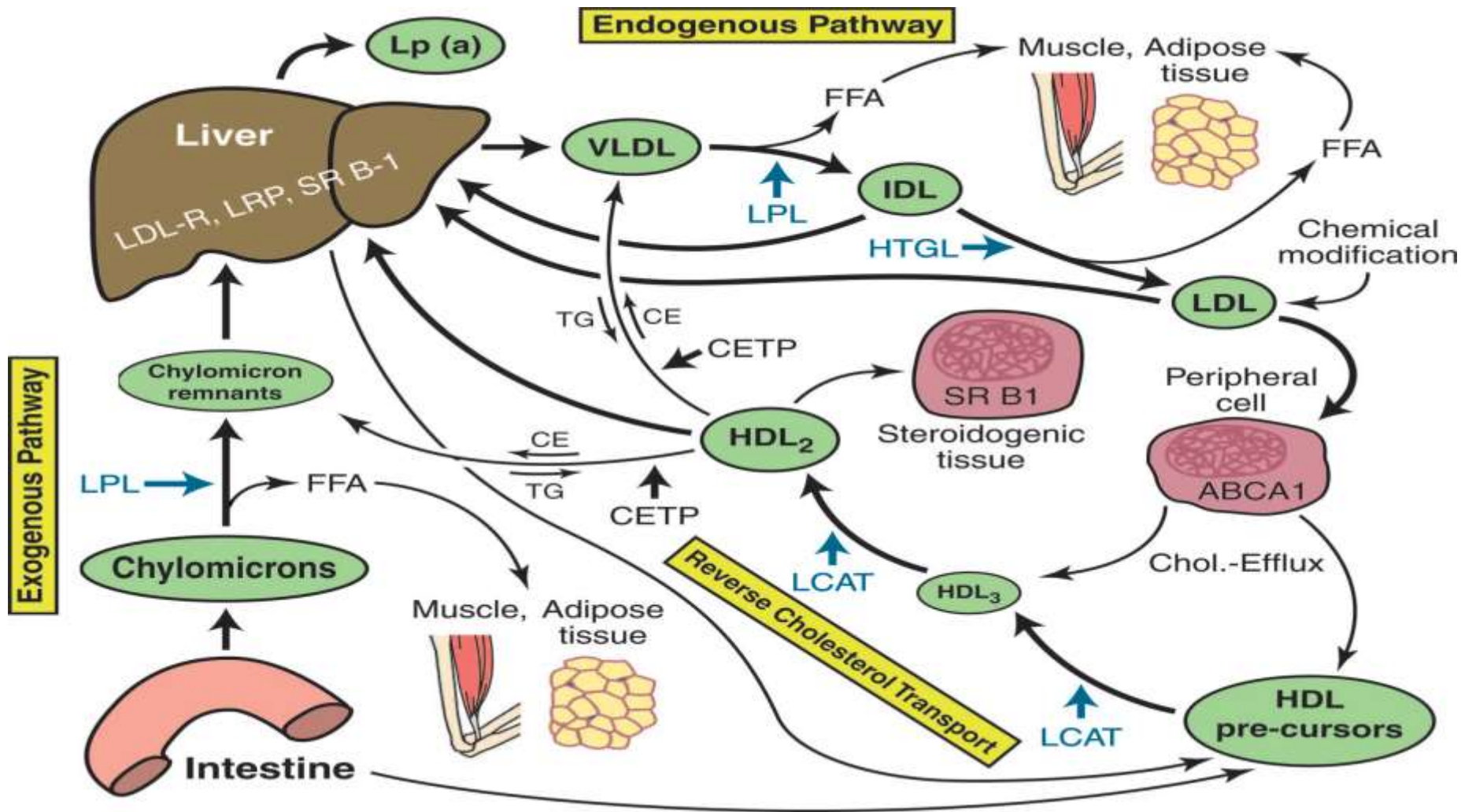


Fig: . Lipoproteins metabolism in exogenous and endogenous pathways
 ABCA1, ATP binding cassette transporter 1 ; CE, cholesterol ester; CETP, cholesteryl ester transfer protein; FFA, free fatty acid; HTGL, hepatic triglyceride lipase; LCAT, lecithin: cholesterol acyltransferase; LPL, lipoprotein lipase; LRP, LDL-R-related protein; SRB1, scavenger receptor B1.

Lipoprotein Profile

- Cholesterol levels are measured in milligrams (mg) of cholesterol per deciliter (dL) of blood.
- Desirable or optimal levels for adults are:
 - Total cholesterol: Less than 200 mg/dL.
 - Low Density Lipoprotein (LDL) cholesterol: Less than 100 mg/dL
 - High Density Lipoprotein (HDL): 40 mg/dL or higher.
 - Triglycerides: Less than 150 mg/dL.
- The National Cholesterol Education Program recommends that healthy adults over 20 years of age have their cholesterol levels checked once every 5 years

NB: “Read on Frederickson’s classification of hyperlipoproteinemias”

Thank You!!